

EUREKA ROADHOUSE, AK, USA

WMO: 702715

Lat: 61.939N

Lon: 147.167W

Elev: 3289

StdP: 13.03

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 46407

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-17.3	-28.7	1.2	-13.9	-26.7	1.4	-13.8	18.7	16.6	16.6	16.1	4.6	50	0.818

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	69.5	53.8	64.4	51.6	62.5	50.4	55.3	65.3	53.2	62.1	51.4	59.5	6.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
50.4	61.2	57.5	48.3	56.5	55.7	46.4	52.7	53.8	24.6	66.2	23.2	62.3	22.1	59.5	63.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.6	14.2	12.3	-24.4	74.3	3.3	3.9	-26.8	77.1	-28.7	79.4	-30.6	81.6	-33.0	84.4	
			-23.8	57.0	3.6	2.9	-26.4	59.1	-28.5	60.7	-30.5	62.3	-33.0	64.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	3.1	7.7	11.4	25.5	39.2	49.0	52.0	49.1	39.7	24.8	9.1	5.0
	DBStd	19.98	12.09	11.17	10.29	7.94	7.05	5.82	4.57	5.21	6.02	7.93	10.32	10.61
	HDD50	8824	1455	1184	1197	735	343	87	30	80	311	782	1226	1394
	HDD65	14090	1920	1604	1662	1185	801	480	403	494	759	1247	1676	1859
	CDD50	211	0	0	0	0	7	58	93	51	2	0	0	0
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH74	11	0	0	0	0	0	9	1	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.9	3.5	3.8	4.8	5.2	6.5	7.2	6.8	5.8	5.2	3.6	3.4	3.1
Precipitation	PrecAvg	50.5	3.7	4.1	3.1	2.7	2.7	3.0	4.3	5.3	6.5	5.6	4.1	4.8
	PrecMax	63.4	11.1	10.6	6.0	5.3	6.3	4.7	6.9	14.2	11.3	13.9	11.6	9.4
	PrecMin	34.1	0.2	0.0	0.4	0.6	0.8	0.8	1.9	1.0	1.5	2.5	0.0	0.7
	PrecStd	6.5	2.6	2.3	2.0	1.2	1.5	0.9	1.4	3.1	2.7	3.0	3.0	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	30.4	36.5	48.3	65.7	74.5	72.0	72.6	59.0	47.7	34.0	31.7
		MCWB	29.4	27.4	31.8	39.3	48.9	55.1	55.7	56.5	47.0	40.9	30.8	30.3
	2%	DB	27.4	27.9	32.1	43.0	61.1	69.0	67.8	66.1	54.3	39.3	29.9	27.5
		MCWB	25.6	25.8	28.6	34.5	45.6	52.5	53.9	53.6	44.5	35.3	28.9	26.1
	5%	DB	23.0	25.3	27.9	38.9	55.3	63.8	64.1	62.6	51.8	36.6	27.2	23.4
		MCWB	21.4	22.8	25.0	32.4	43.1	50.3	52.3	51.7	43.4	33.1	25.9	21.9
	10%	DB	18.9	21.3	25.1	36.5	52.1	60.7	61.4	57.2	48.1	34.3	23.4	19.3
		MCWB	17.9	19.9	21.9	30.8	41.4	48.4	50.8	48.9	41.7	31.5	21.8	18.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	29.7	32.2	39.4	49.3	58.4	57.3	57.5	47.9	41.6	31.4	30.3
		MCDB	31.4	31.3	36.2	47.6	63.5	69.9	67.9	69.8	56.2	46.7	33.0	31.0
	2%	WB	25.9	26.3	28.8	35.6	46.3	54.2	55.1	54.8	45.4	36.6	28.5	26.3
		MCDB	27.4	27.6	31.4	42.2	59.6	64.7	65.0	64.4	52.1	39.0	29.2	27.2
	5%	WB	21.7	23.7	25.5	33.1	43.9	51.3	53.4	52.3	43.9	33.5	25.9	22.3
		MCDB	23.2	26.4	27.9	38.8	54.6	61.3	62.0	60.1	49.6	35.8	27.2	23.5
	10%	WB	18.3	20.7	22.4	31.3	41.5	49.2	51.9	50.3	42.3	31.8	22.2	19.0
		MCDB	19.1	22.2	25.3	36.2	50.9	58.5	60.0	56.7	47.5	34.1	23.7	19.6
Mean Daily Temperature Range	5% DB	MDBR	10.8	10.7	12.5	13.6	15.4	16.4	15.0	13.6	11.9	9.1	9.7	10.6
		MCDBR	11.9	12.8	13.5	14.3	21.6	21.4	19.1	18.6	15.5	11.3	10.2	12.5
		MCWBR	11.1	11.4	10.9	9.8	11.3	9.8	9.1	9.2	8.8	8.0	9.4	12.0
	5% WB	MCDBR	12.1	12.3	13.1	13.6	20.6	19.8	17.5	17.4	14.2	9.9	9.9	12.2
		MCWBR	11.6	11.2	10.7	9.4	11.0	9.4	8.6	8.9	8.5	7.5	9.4	12.0
Clear-Sky Solar Irradiance	taub	0.213	0.279	0.285	0.334	0.366	0.355	0.357	0.354	0.306	0.273	0.224	0.202	
	taud	1.983	2.030	2.102	2.024	2.147	2.315	2.333	2.343	2.435	2.232	1.939	1.838	
	Ebn at Noon	150	208	260	267	269	275	271	262	257	215	134	82	
	Edh at Noon	12	24	32	43	42	36	35	31	23	19	12	8	
All-Sky Solar Radiation	RadAvg	93	340	899	1520	1782	1791	1627	1255	807	374	132	46	
	RadStd	10	31	61	83	107	100	78	84	75	35	17	4	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page